

AI Usage Report

University ERP · Advanced Software Engineering · UCSC

1. Introduction

This report documents every interaction with AI tools during the development of the University ERP microservice assignment. It covers architecture design, document generation, diagram creation, and prototype code scaffolding. Each entry records the tool used, the exact task, a summary of the AI output, human corrections applied, how the output was validated, and any hallucinations or errors observed.

2. AI Usage Log

Entry 1 — Analyse architecture diagram (PNG)

Tool used	Claude claude-sonnet-4-6 via Claude Code CLI
Task / purpose	Read the revised_architecture_nodejs_mongodb.png image and extract the full architecture: all 8 services, ports, databases, libraries, NGINX routing, and Kafka event bus to understand what needs to be built.
Prompt used	"Read this revised_architecture_nodejs_mongodb.png and according to that I want to build set of node applications. This is actually microservice project."
AI output (summary)	Claude identified all 8 services (Auth :3007, Student :3001, Course :3002, Exam :3003, Result :3004, Transcript :3005, Notification :3006, Reporting :3008), their MongoDB databases, library dependencies per service (mongoose, axios, kafkajs, pdfkit, bcrypt, jsonwebtoken), NGINX path routing table, and three Kafka events (student.created , exam.registered , result.published). It proposed a monorepo structure under services/.
Human corrections	Previously, Redpanda or Kafka was used as the event-driven communication technology. Redpanda has since been removed from the architecture.
Validation method	Done analysis through Claude Code whether we are using Redpanda as event

	driven technology or not. Make sure all the database deployed in one cluster.
Mistakes / hallucinations	None
Ethical / privacy concerns	No sensitive data in prompt. The PNG is an architecture diagram with no personal information.
Reflection	Accurately extract diagram and proceed development according to the given architecture.

Entry 2 — Generate Service Identification Table

Tool used	Claude claude-sonnet-4-6 via Claude Code CLI
Task / purpose	Generate Service Identification Table_Service_Identification_Table.docx covering all 8 services with port, business capability, owned data, database, and communication type. Also include inter-service communication tables and technology stack.
Prompt used	"yes start" (after confirming the plan to begin with Service Identification Table and the Prototype prototype together)
AI output (summary)	Generated a 5-section DOCX: (1) purpose, (2) service identification table with all 8 services, (3) service boundary rules, (4) sync REST and async Kafka communication tables, (5) technology stack table with rationale. Covered all 8 services with correct ports, DB names, and library lists.
Human corrections	Incorrect information not found during the generation.
Validation method	Check architecture diagram when validating the service table.
Mistakes / hallucinations	No mistakes
Ethical / privacy concerns	No sensitive data.
Reflection	According to the architecture diagram and the development, properly created.

Entry 4 — Scaffold Prototype — Student Service

Tool used	Claude claude-sonnet-4-6 via Claude Code CLI
Task / purpose	Scaffold a complete Node.js/Express Student Service (port 3001) with Mongoose model, JWT middleware, REST routes, Dockerfile, and package.json.
Prompt used	Implied by "yes start" confirmation to build prototype-first (Option 1).
AI output (summary)	Generated: src/index.js (Express app, CORS, /health endpoint), src/config/db.js (Mongoose connect), src/middleware/auth.js (JWT verify),

	src/models/Student.js (name, email, studentId, programme + timestamps), src/routes/students.js (POST /api/students, GET /api/students/:id, GET /api/students), Dockerfile (node:20-alpine), package.json (express, mongoose, jsonwebtoken, cors, dotenv).
Human corrections	According to the requirement student service was implemented.
Validation method	Ran locally through Docker compose.
Mistakes / hallucinations	No mistakes
Ethical / privacy concerns	No sensitive data.
Reflection	According to the architecture diagram and use-case that defined in the requirement and existing system, properly implemented.

Entry 5 — Scaffold Prototype — Exam Service with cross-service call

Tool used	Claude claude-sonnet-4-6 via Claude Code CLI
Task / purpose	Scaffold Exam Service (port 3003) with Exam and Entry Mongoose models, routes including POST /api/exams/:id/entries that calls Student Service internally via axios to verify a student before creating an Entry.
Prompt used	Same session as Student Service scaffolding.
AI output (summary)	Generated: Exam.js model (title, courseId, date, venue), Entry.js model (examId, studentId, studentName, enrolledAt + compound unique index), exams.js routes with 5 endpoints including the cross-service axios call, error handling for 404 (student not found) and 502 (Student Service down), Dockerfile, package.json (adds axios and kafkajs to dependencies).
Human corrections	According to the requirement exam service was implemented.
Validation method	Ran docker-compose up --build and tested the cross-service flow: created a student, then enrolled them in an exam, then attempted enrolment with an invalid studentId to verify 404 responses.
Mistakes / hallucinations	No Mistakes
Ethical / privacy concerns	No sensitive data.
Reflection	According to the architecture diagram and use-case that defined in the requirement and existing system, properly implemented.